Writing Software for TOFs

Programmer Guide

This manual discusses general issues of TOF software.

1.1. TOF - to - Mass Conversion

The time-of-flight t of ions in electrostatic fields is proportional to the square root of their mass per charge m:

$$t = a\sqrt{m}$$

The measured time-of-flight t_m of ions usually includes a time offset between the start of the ions and the start of the acquisition electronics. This time offset t_0 is either made on purpose (to not record the ringing of the ion acceleration pulser) or it is due to signal transition time differences in the electronics.

$$t_m = t_0 + a\sqrt{m}$$

1.2. Mass Calibration

When transforming TOF spectra to mass spectra, the two parameters t_0 and a have to be determined. This is done by solving the system of equation

$$t_1 = t_0 + a\sqrt{m_1}$$
$$t_2 = t_0 + a\sqrt{m_2}$$

where t_1 and t_2 are the time-of-flight of the known masses m_1 and m_2. The system solves to:

$$t_0 = \frac{\sqrt{m_1}t_2 - \sqrt{m_2}t_1}{\sqrt{m_1} - \sqrt{m_2}}$$
$$a = \frac{t_1 - t_0}{\sqrt{m_1}}$$

With those two parameters all TOF-to-mass conversions can be done quite reliably if mass and position of the two calibration peaks are known accurately.
The calibration parameters should be saved in a file for the next use of the software. However, the software should include a routine where the times t_1 and t_1 of two known masses m_1 and m_2 can be assigned. This can be done automatically or by user input.

1.3. TOF Resolving Power

For TOFs the resolving power is defined as:

$$R(m) = \frac{m}{\Delta m}$$

or, when expressed in time space as:

$$R(t) = \frac{t_0}{2 \cdot \Delta t}$$

where t_0 is the "position" of the peak, which is best defined as the peak centroid, however, it is also possible to use the peak maximum.

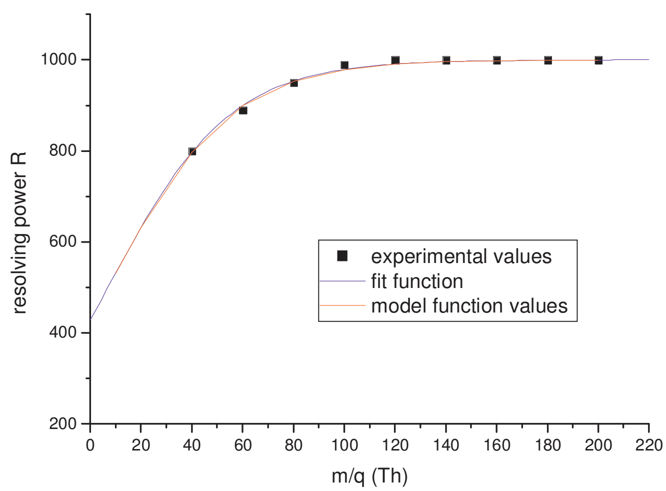
1.4. Low Mass Resolving Power measured with ADC

For Tofwerks' TOF the mass resolving power R is approximately constant over the whole mass range. This, however, is not true for low masses (1 to 100 Th). In this range the single ion peak represents a significant portion of the overall peak width measured with an ADC. For the peak integration we need the following approximation for the mass dependency of the resolving power R:

$$R(m) = \frac{-R_0}{1 + e^{(m-m_0)/dm}} + R_0$$

For the C-TOF operated with ADC data acquisition the following parameters have been determined:

R_0	=	1000	nominal resolving power
m_0	=	7	mass at which resolving power is $R_0/2$
dm	=	24.213	slope parameter
m	=		mass (more precisely: mass/charge)



data by Peter DeCarlo